Research Report:

Personalized AI for Parkinson Patients

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Abstract

Short clinical consultations leave limited time for people with Parkinson's disease to describe weeks of motor fluctuation. Wrist-worn accelerometers can record movement continuously, but raw sensor data are difficult for people to interpret and provide limited semantic context for a large language model (LLM). This report examines how data representation and prompt structure shape LLM-generated patient-facing narratives. Three multi-day, free-living wrist-accelerometry recordings were analysed, totalling 307.3 hours at 100 Hz. Each recording was presented to the same LLM under three conditions: raw tri-axial samples, an unsegmented signal vector magnitude (SVM) series, and a structured Multi-Symptom Summary Condition (MSSC). The raw condition led the model to explain why the input was insufficient for quantitative interpretation. The SVM condition supported broad descriptions of activity variation, but also generated unsupported clock times and general descriptions of calm movement. The MSSC condition enabled close verbal correspondence with the supplied algorithmic percentages, while the prompt introduced medication-related questions that were not supported by medication timestamps. The findings suggest that structured summaries give the LLM more explicit semantic context, but that prompt instructions can still introduce clinical framing beyond the available telemetry.

Keywords: Parkinson's disease; wearable sensors; large language models; prompt engineering; patient-facing narratives; human-computer interaction; digital health.

1 Introduction and Motivation

People with Parkinson's disease may experience tremor, slowness, and involuntary movement that vary across the day. A neurologist may see only a short part of this experience during a routine consultation. The patient must therefore recall events from previous days or weeks, while the clinician must interpret this account together with the examination performed during the visit.

Wrist-worn accelerometers offer one way to capture movement during daily life. However, raw accelerometry is a high-volume numerical time series. It is not designed as a patient-facing explanation, nor as a direct input for a text-generation model. The practical question is therefore not whether a wrist sensor can replace clinical assessment. It is how sensor-derived information can be represented so that it can support understandable reflection and discussion.

Large language models can turn structured information into fluent text. In health settings, this creates an opportunity to translate technical summaries into accessible language. It also

creates a risk: a coherent narrative may include details, causes, or clinical interpretations that are not contained in the input. This report studies that interaction between sensor-data representation, prompt structure, and generated narrative.

The work does not evaluate a medical device or validate a clinical diagnostic algorithm. The symptom-oriented signal-processing layer is used to construct different forms of input for the LLM. The primary object of study is how the LLM responds when it receives progressively more structured context.

2 Background

2.1 Wearable sensing in Parkinson's disease

Wearable sensors have been used to quantify movement-related features in Parkinson's disease. Tremor is clinically defined as an involuntary, rhythmic, oscillatory movement, classified according to clinical phenomenology and aetiology [1]. Salarian et al. demonstrated ambulatory inertial sensing for tremor and bradykinesia quantification [2]. Griffiths et al. described continuous wrist-based measures of bradykinesia and dyskinesia [3]. Cole et al. and Tsipouras et al. investigated automated detection of tremor and dyskinesia using wearable signals and computational features [4–6]. Where clinical comparison was performed, sensor-derived measures were commonly related to assessments such as the MDS-UPDRS [7].

These studies motivate the use of signal processing to derive movement-related descriptors. They do not validate the exact thresholds used in this report, nor do they establish that an LLM-generated narrative is clinically valid.

2.2 LLMs as a communication layer

LLMs can generate concise, readable text from structured information. In health communication, this capability may help translate technical records into language that patients can understand. However, patient-facing text requires particular caution: fluent wording can make unsupported content appear credible. LLM-based health communication has therefore been associated with concerns about omissions, hallucinations, and the need for human oversight [8].

The use of LLMs with wearable data is also emerging. Wearable signals are high-volume, continuous, and difficult to interpret without preprocessing. Reviews of LLMs for wearable sensing identify data representation, interpretability, and integration with conventional signal-processing methods as central challenges [9]. This report adopts an HCI perspective: it examines how the interface between data representation and prompt instructions shapes the narrative that the model produces.

3 Research Goal

The broader objective is to explore how wearable data can be transformed into language that may support reflection and consultation preparation without presenting algorithmic outputs as medical facts.

The report addresses the following research question:

> *How does a large language model summarise long-term wearable sensor data across different representations, and can it generate a clinically efficient and safe conversational narrative for Parkinson's disease patients?*

4 Methodology

4.1 Dataset and recording selection

The dataset consisted of three anonymised CSV recordings of triaxial wrist accelerometry collected during unconstrained daily activity. Each file contained a timestamp and acceleration values for the x, y, and z axes. The sensor was worn on the left wrist. Sampling at 100 Hz corresponds to one sample every 10 ms.

The three recordings were selected to provide contrasting algorithmic movement profiles. Recording 30837 showed a predominantly stable or low-activity pattern. Recording 35637 showed mixed bradykinesia- and dyskinesia-criterion intervals. Recording 33288 showed a highly active pattern with many windows satisfying the dyskinesia criterion. These descriptions refer only to the outputs of the implemented signal-processing pipeline; they are not independently validated clinical phenotypes.

No medication diary, clinician annotation, or independently labelled symptom timeline was available. Consequently, the signal-processing outputs are used as algorithmic descriptors for the LLM conditions, not as confirmed clinical measurements.

Table 1: Free-living wrist-accelerometry recordings used in the study.

Participant ID	Sampling rate	Samples	Duration (h)	Sensor location
30837	100 Hz	44,675,037	124.10	Left wrist
35637	100 Hz	35,911,520	99.75	Left wrist
33288	100 Hz	30,039,040	83.44	Left wrist
Total	–	110,625,597	307.29	–

4.2 Signal processing and symptom-oriented descriptors

The signal-processing layer was used to construct structured input, not to establish a diagnostic system. Acceleration was filtered independently per axis with a Butterworth high-pass filter with cutoff frequency $f_c = 0.5$ Hz. The filter was intended to attenuate gravitational and very-low-frequency orientation components.

Signals were segmented into 4-second windows with 50% overlap. For tremor analysis, the power spectral density was estimated with Welch's method for each signed axis and then summed:

$$PSD_{\text{total}}(f) = PSD_x(f) + PSD_y(f) + PSD_z(f).$$

The dominant frequency was identified from the summed spectra. The root-mean-square value for a window of N samples was:

$$RMS = \sqrt{\frac{1}{N}\sum_{i=1}^{N} s_i^2}.$$

Three independent, rule-based symptom criteria were then evaluated:

- **Resting tremor:** a dominant spectral peak in the operational 4–6 Hz band, combined with an RMS-based resting-activity condition and temporal persistence. The band was motivated by the approximate 4–7 Hz range often associated with parkinsonian resting tremor [1]. Exact RMS and persistence values were implementation parameters.
- **Bradykinesia:** active windows were analysed in the 0.2–4.0 Hz movement band using dominant frequency, amplitude decrement, and low-variation intervals. These features

were motivated by descriptions of slowness, decrement, hesitation, and interruption in the MDS-UPDRS and wearable-sensor work [2, 3, 7]. Thresholds were operational choices.

- **Dyskinesia:** elevated RMS relative to an individual baseline, rejection of windows dominated by rhythmic gait-like activity between 1.0 and 2.5 Hz, normalised spectral entropy in the 0.5–4.0 Hz band, and temporal persistence. These features were informed by automated dyskinesia research [5, 6]. Entropy, gait, RMS, and persistence thresholds were implementation parameters.

The use of clinical and wearable literature informed the choice of features and broad frequency ranges. It does not imply that the numerical thresholds reproduce a validated protocol from any single study.

4.3 Input representations

Three representations were constructed from each recording.

1. **Raw condition:** a representative 3.5-second excerpt containing 350 timestamped raw x, y, and z samples, including gravity.
2. **SVM condition:** a full-recording signal vector magnitude representation. The dynamic magnitude was defined as:

$$SVM_{\text{dyn}}(t) = \sqrt{x_{\text{dyn}}(t)^2 + y_{\text{dyn}}(t)^2 + z_{\text{dyn}}(t)^2}.$$

   Here, SVM denotes signal vector magnitude, not support vector machine. This condition retained a coarse view of activity variation without explicit symptom labels.
3. **Multi-Symptom Summary Condition (MSSC):** a structured summary with four descriptors: windows with no positive symptom criterion, tremor-criterion windows, bradykinesia-criterion windows, and dyskinesia-criterion windows.

For each symptom criterion s, the percentage was calculated as:

$$\text{Metric Percentage}_s = \left(\frac{N_{\text{windows satisfying criterion } s}}{N_{\text{total valid windows}}} \right) \times 100.$$

The symptom-free category included windows in which none of the three detectors returned a positive criterion. Positive symptom criteria were evaluated independently and could co-occur in the same window. Therefore, positive criterion percentages can overlap and their sum with the symptom-free percentage may exceed 100%.

4.4 Prompt design

The prompts requested an empathetic, non-alarmist tone and avoided engineering terms such as "RMS", "Hz", "g", and raw decimal values. The conversational output followed three parts:

1. *Your Week in a Few Words*: a broad summary;
2. *For Reflection*: questions intended to support episodic recall;
3. *What to Bring to the Doctor*: a short first-person statement.

The structure was selected to separate description, reflection, and preparation for discussion. It was not designed to provide diagnosis or treatment advice.

The following excerpt shows the experimental MSSC prompt structure:

You are an empathetic health assistant for people with Parkinson's disease. Transcribe the watch monitoring summary into a structured and positive dialogue. Time with calm and natural movements: approximately `{tempo_estavel}`*% of the monitored period. Time with slowness or pauses in gestures: approximately* `{perc_brady}`*%. Time with brief episodes of resting tremor: approximately* `{perc_tremor}`*%. Time with brief moments of involuntary movements: approximately* `{perc_dysk}`*%. Present the response in three sections: (1) Your Week in a Few Words; (2) For Reflection; and (3) What to Bring to the Doctor. In the reflection section, ask whether slowness was noticed closer to dose wear-off and whether agitation appeared at peak effect.*

The medication-related questions were explicitly required by the experimental prompt. They were not inferred from medication timestamps, because no such data were supplied.

4.5 Experimental design and output analysis

The study used three recordings under three representation conditions, resulting in $3 \times 3 = 9$ LLM inference runs. The model was `openai/gpt-oss-120b`, accessed through Groq. Temperature was fixed at 0.2. The model, general patient-facing task, and output structure were held constant where possible; the representation-specific input and instructions differed by condition.

Generated outputs were analysed through qualitative analytical inspection. The inspection considered numerical correspondence with supplied values, textual coherence, adherence to the requested structure, and whether temporal or clinical details were supported by the input. No neurologist panel, patient panel, inter-rater reliability measure, or formal clinical usability assessment was used. The analysis is therefore exploratory and descriptive.

5 Results

Table 2: Observed characteristics of generated outputs across data representations.

Dimension	Raw input	SVM input	MSSC input
Data description	The model explained that reliable quantitative signal analysis could not be performed from the textual sample alone.	The model described broad periods of relative activity and inactivity.	The model verbalised the supplied criterion percentages.
Narrative form	The output mainly described the limitations of the input.	The output followed a conversational format and described broad movement patterns.	The output followed the requested three-part format: summary, reflection, and doctor-oriented statement.
Additional content	The model included qualitative hypotheses while emphasising uncertainty.	The model included weekday and clock-time references not present in the input.	The model included medication-related questions requested by the prompt.

5.1 Raw condition

In the Raw condition, the model received a short tri-axial numerical excerpt. It explained that the text table did not provide enough information for reliable filtering, spectral analysis, or numerical feature extraction. The output focused on limitations of the representation and avoided a definitive clinical classification.

The response also offered qualitative possibilities, such as low activity, possible movement adjustment, or uncertain tremor-like variation. These possibilities were framed as tentative because the input did not contain computed spectral or temporal features.

5.2 SVM condition

In the SVM condition, the model received a magnitude-based representation of the full recording. The generated text described broad changes in movement intensity and referred to relatively quieter and more active periods. Thus, the SVM input supported a macro-level account of activity and inactivity across the monitored period.

The model also generated specific calendar references, such as "around 11 a.m. on Monday" and "around 2 p.m. on Tuesday". These times were not supplied as identified activity events in the input representation. For recording 33288, the narrative described the monitored period as mostly calm and referred to good motor control.

5.3 MSSC condition

In the MSSC condition, the model received a structured summary of no-positive-criterion windows and the three symptom-criterion percentages. The generated outputs followed the requested three-part format.

For recording 33288, the narrative verbalised approximately 63% of windows satisfying the dyskinesia criterion and approximately 24% with no positive criterion. The output therefore reflected the supplied algorithmic values in natural language. This observation concerns numerical correspondence between the MSSC input and the generated text; it does not validate the symptom criteria.

The model also produced questions about dose wear-off and peak-dose effects. These questions followed the explicit prompt instruction and were not based on medication timestamps in the sensor data.

Table 3: Pipeline-derived criterion percentages used as MSSC input.

Participant ID	No positive criterion (%)	Tremor criterion (%)	Bradykinesia criterion (%)	Dyskinesia criterion (%
30837	85.0%	5.99%	8.26%	1.47%
35637	42.9%	1.60%	35.87%	32.53%
33288	24.4%	1.34%	31.39%	63.13%

6 Discussion

6.1 Progressive context and generated narratives

The three conditions represent increasing semantic context. The Raw condition preserved the closest representation to the measurement, but supplied little explicit meaning for narrative generation. The model therefore concentrated on explaining why further processing would be needed.

The SVM condition introduced an intermediate abstraction. It reduced the three axes to a single activity magnitude and enabled broad descriptions of quieter and more active periods. This was useful for macro-level variation. However, it did not explicitly provide movement direction, episode boundaries, dominant frequencies, or symptom-criterion labels. The model therefore generated a broad activity narrative rather than a symptom-oriented summary.

The MSSC condition provided the most semantic structure. It explicitly named no-positive-criterion windows and the three symptom criteria, together with their percentages. This made it possible for the model to verbalise the supplied numbers more directly. The trade-off is that

the model inherited the assumptions and limitations of the heuristic signal-processing criteria. The MSSC improves the accessibility of these descriptors; it does not make the upstream labels clinically validated.

6.2 Prompt-induced clinical framing

The medication-related questions illustrate an HCI risk that arises from prompt design. The telemetry contained no medication timestamps, dose information, or self-reported experience of wearing-off. Nevertheless, the prompt instructed the model to ask about wear-off and peak-dose effects.

The model did not independently discover these medication patterns. It followed the prompt. However, a patient-facing question can still suggest a causal explanation, even when written as a question. This distinction matters because users may interpret a plausible clinical frame as evidence from their data.

Data-grounded observations and reflective scaffolding should therefore be kept separate. A future prompt could ask a neutral question such as "Did you notice any pattern around medication times?" only when medication timing is available or when the question is clearly labelled as a topic for personal reflection rather than a sensor-derived finding.

6.3 Relation to prior work and limitations

Prior wearable work supports the need for preprocessing and feature extraction before movement-related interpretation [2, 3]. Recent work on LLMs and wearable sensing also highlights the importance of data modelling, representation, and interpretability [9]. This report extends that concern to patient-facing narrative generation.

The study has several limitations. It used three recordings and one LLM, with one run per participant-condition combination. The narrative outputs were assessed by qualitative inspection rather than by neurologists, patients, or independent raters. The sensor data contained no medication diary and no clinical ground truth. The report therefore describes observed behaviours of a specific model under specific representations and prompts; it does not measure clinical validity, safety, usability, or patient outcomes.

7 Conclusion

This report examined how a large language model generated patient-facing narratives from three levels of wearable-sensor representation: raw tri-axial samples, signal vector magnitude, and a structured Multi-Symptom Summary.

The Raw condition led the model to explain the limits of interpreting a short numerical excerpt. The SVM condition supported broad descriptions of activity variation, but also produced unsupported calendar details and did not provide explicit symptom-oriented structure. The MSSC condition enabled closer verbal correspondence with supplied algorithmic percentages because it provided explicit descriptors and proportions.

The findings suggest that data representation and prompt structure jointly shape the generated narrative. Structured metrics may improve the semantic context available to the model, but they can also carry forward assumptions from the upstream heuristic detectors. Prompt instructions can add medical framing that is absent from the telemetry.

A separated architecture is therefore appropriate:

Sensors → Deterministic DSP → Structured Metrics → Constrained Prompts → Patient Narratives.

Future Work

Future work should evaluate this architecture with larger and more diverse cohorts, repeated runs, multiple LLMs, and controlled prompt variants. Neurologists and people with Parkinson's disease should assess the numerical fidelity, clarity, perceived usefulness, trust, and potential risks of generated narratives. A formal usability study should assess whether the format supports reflection and consultation preparation without increasing confusion or anxiety.

Synchronized medication diaries should be integrated so that medication-related questions can be grounded in actual dose times. Future prompts should also include explicit constraints against diagnosis, invented timestamps, unsupported subjective experiences, medication inference without medication data, and treatment recommendations. Independent clinical annotations would further allow the signal-processing outputs to be evaluated separately from the language-generation layer.

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